

2.7 The periodic table — groups 2 and 7

During this topic there will be the opportunity to carry out a number of internal assessment activities. Please see *Appendix 9* for more details.

Students will be assessed on their ability to:

**1 Properties
down group 2**

- a explain the trend in the first ionization energy down group 2
- b recall the reaction of the elements in group 2 with oxygen, chlorine and water
- c recall the reactions of the oxides of group 2 elements with water and dilute acid, and their hydroxides with dilute acid
- d recall the trends in solubility of the hydroxides and sulfates of group 2 elements
- e recall the trends in thermal stability of the nitrates and the carbonates of the elements in groups 1 and 2 and explain these in terms of size and charge of the cations involved
- f recall the characteristic flame colours formed by group 1 and 2 compounds and explain their origin in terms of electron transitions
- g describe and carry out the following:
 - i experiments to study the thermal decomposition of group 1 and 2 nitrates and carbonates
 - ii flame tests on compounds of group 1 and 2
 - iii simple acid-base titrations using a range of indicators, acids and alkalis, to calculate solution concentrations in g dm^{-3} and mol dm^{-3} , eg measuring the residual alkali present after skinning fruit with potassium hydroxide
- h demonstrate an understanding of how to minimise the sources of measurement uncertainty in volumetric analysis and estimate the overall uncertainty in the calculated result.

**2 Inorganic
chemistry
of group 7
(limited to
chlorine,
bromine and
iodine)**

- a recall the characteristic physical properties of the elements limited to the appearance of solutions of the elements in water and hydrocarbon solvents
- b describe and carry out the following chemical reactions of halogens:
 - i oxidation reactions with metal and non-metallic elements and ions such as iron(II) and iron(III) ions in solution
 - ii disproportionation reactions with cold and hot alkali, eg hot potassium hydroxide with iodine to produce potassium iodate(V)